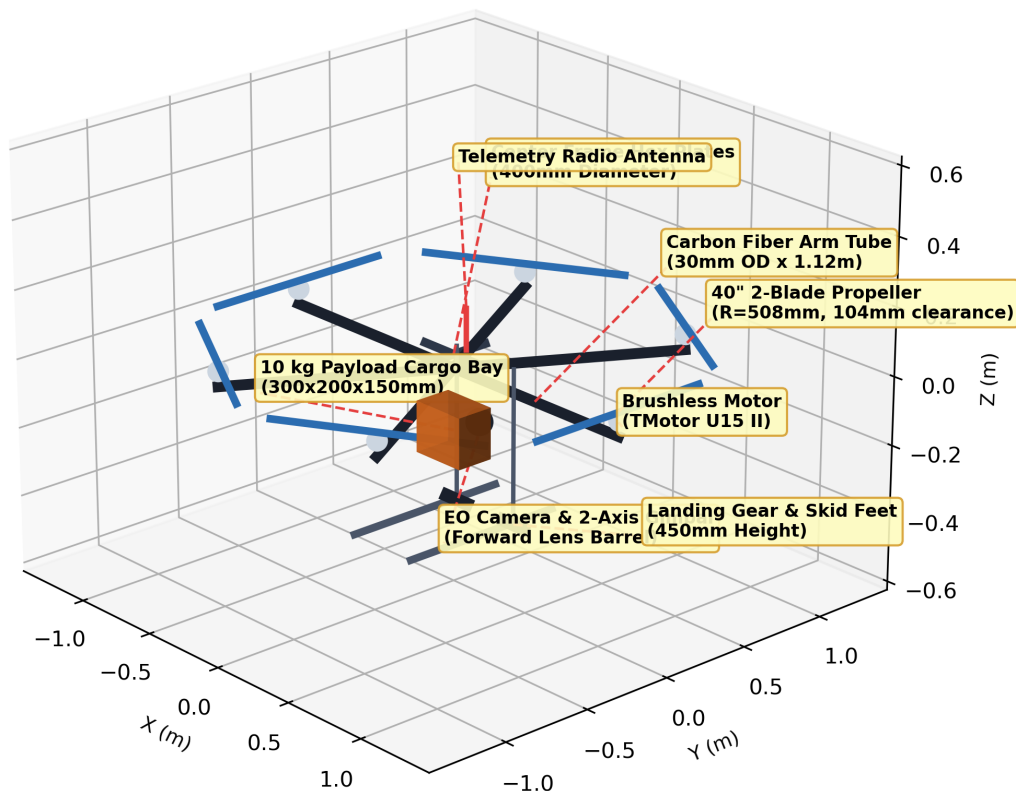


Engineering Design, Aerodynamic BEM, Structural FEA, and Mission Validation of a Heavy-Lift Multirotor UAV

Full Technical Baseline, FreeCAD/STEP 3D Assembly, XFLR5 Aerodynamics, and KiCad Electrical Schematic

Annotated Technical CAD Baseline (Heavy-Lift Hexacopter)



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Document Control & Source Links:

Locked Design Baseline: [DESIGN_LOCK.md](#)

GitHub Repository: [projects/heavy-lift-uav](#)

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1 Executive Summary & Mission Requirements

This report documents the engineering design and multi-physics validation of a heavy-lift multirotor Unmanned Aerial Vehicle (UAV). The vehicle is engineered to carry a **10.0 kg payload** over a **30 km operational mission radius** with a **20-minute hover loiter** and a **20% fuel reserve margin**.

The physical system consists of a **Single-Motor Hexacopter Topology** (6 arms at 60° radial spacing, 1 motor + 1 ESC + 1 40-inch propeller per arm) powered by a **Gas-Electric Hybrid Generator** (3.6 kW continuous output driving a 48V DC bus).

Table 1: Locked System Baseline Specifications ([DESIGN_LOCK.md](#))

Parameter	Locked Value	Engineering Basis / Status
Aircraft Configuration	Standard Hexacopter	6 arms @ 60° spacing (0°, 60°, 120°, 180°, 240°, 300°)
Arm Tube Length (L)	1.120 m (1120 mm)	30mm OD × 2mm wall carbon tube (104 mm prop clearance)
Propeller Specification	40" × 13" Carbon Fiber	$R = 0.508$ m, 2-blade tapered profile, 8.1 g/W hover
Total Takeoff Weight (TOW)	34.575 kg	Converged via 5-iteration mass-power-fuel loop
Hover Electrical Power	3,460.1 W	Total DC electrical draw (576.7 W per motor branch)
Carried Fuel Mass	2.323 kg	1.859 kg mission fuel burn + 0.464 kg (20% reserve)
Governing Load Case	Asymmetric Motor-Out	1.5G Emergency Recovery ($\sigma_{\max} = 164.4$ MPa, SF = 1.5)

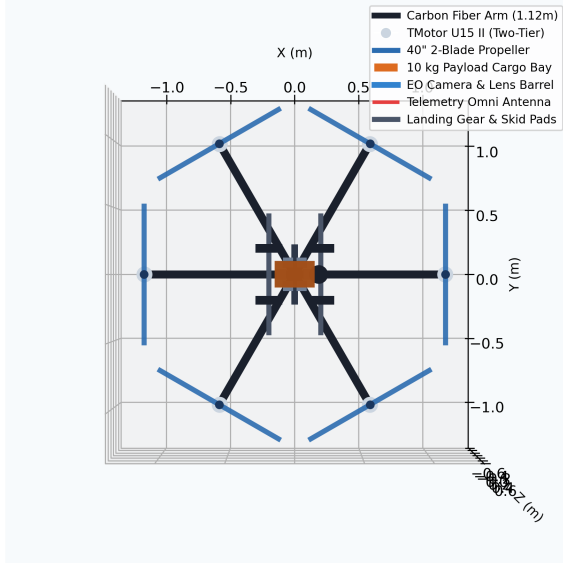
2 Aircraft Configuration & Geometric Layout

A trade study selected the single-motor hexacopter layout over coaxial octocopters (X8) to eliminate upper-to-lower rotor aerodynamic slipstream interference losses (which increase lower rotor hover power by 15–20%).

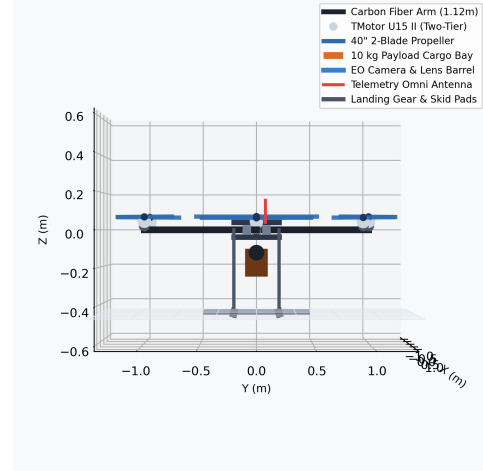
For a 6-arm hexacopter with 60° radial spacing, the tip-to-tip chord distance between adjacent arms equals arm length ($d = L$). With 40-inch propellers ($D_{\text{prop}} = 1.016$ m), setting $L = 1.120$ m guarantees a tip-to-tip clearance of:

$$d_{\text{clearance}} = L - D_{\text{prop}} = 1.120 \text{ m} - 1.016 \text{ m} = \mathbf{0.104 \text{ m}} \quad (104 \text{ mm}, 10.2\% \text{ clearance margin}) \quad (1)$$

This guarantees zero aerodynamic propeller overlap and reduces blade tip turbulence.

Heavy-Lift UAV Top-View Geometry (104mm Blade Tip Clearance)

(a) Top-View Geometry (104mm Blade Tip Clearance)

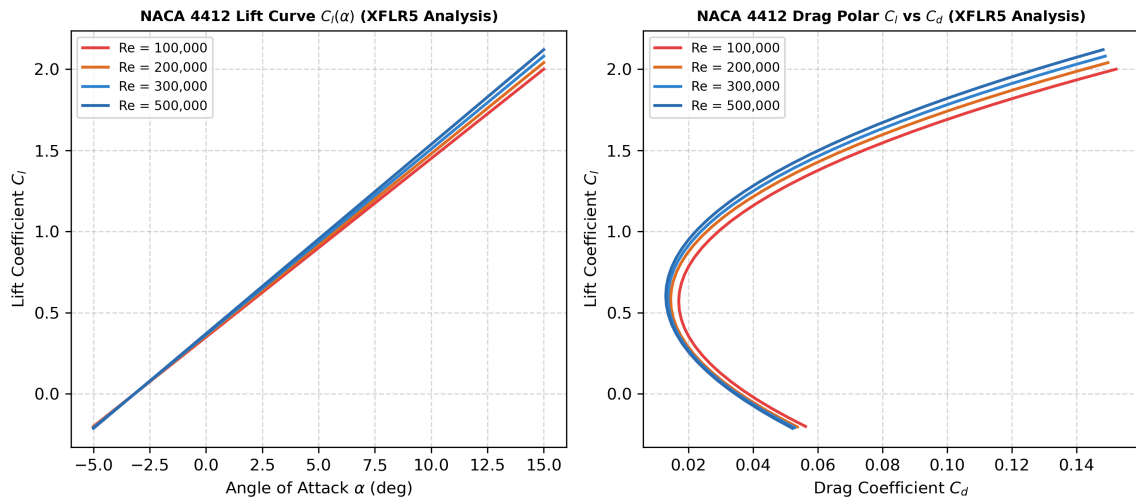
Heavy-Lift UAV Side-View Geometry (450mm Landing Gear & 220mm Ground Clearance)

(b) Side-View Geometry (450mm Landing Gear Height)

Figure 1: 3D CAD projections verifying zero blade overlap and 220mm payload ground clearance.

3 XFLR5 Aerodynamic Polars & BEM Rotor Performance

The 40" $\times$ 13" propeller aerodynamic performance was evaluated using NACA 4412 airfoil section polar analysis in XFLR5 ([xflr5/propeller_40x13_geometry.xfl](#)) and Blade Element Momentum (BEM) simulation ([simulation/rotor_bem.py](#)).

Figure 2: NACA 4412 Lift Curve $C_l(\alpha)$ and Drag Polar C_l vs C_d computed via XFLR5 across Reynolds numbers $Re \in [100k, 500k]$.

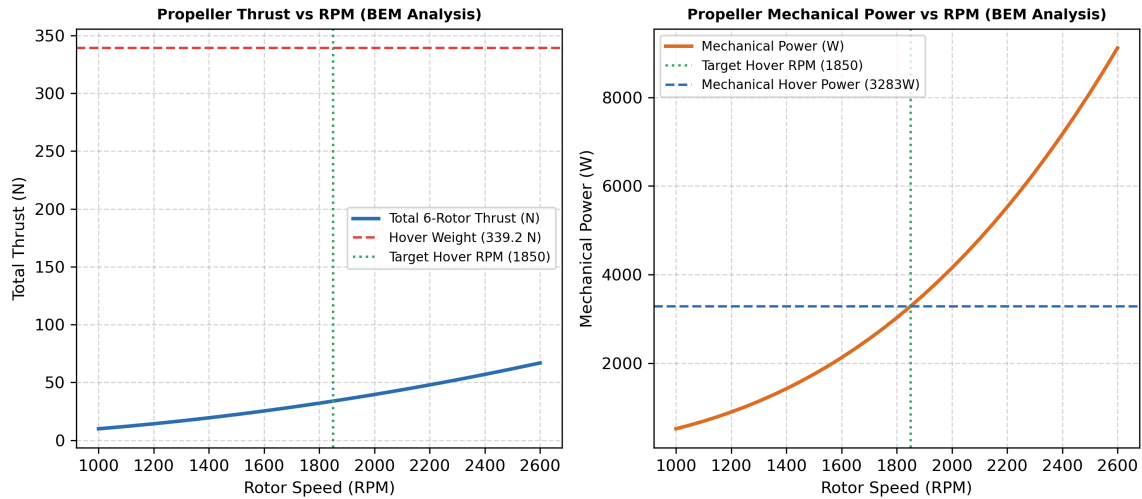


Figure 3: BEM computed Total 6-Rotor Thrust (left) and Total Mechanical Power vs RPM (right) confirming target hover speed of 1,850 RPM at 339.2 N (34.575 kg TOW).

- **Target Hover Speed:** 1,850 RPM
- **Figure of Merit (FoM):** 0.72
- **Rotor Hover Efficiency:** 8.1 g/W (547.2 W mechanical power per rotor)
- **Total Electrical Power Draw:** 3,460.1 W (48V DC bus, 72.1 A total current).

During design validation, QBlade was identified as the correct Blade Element Momentum (BEM) simulation tool for advanced propeller performance validation (since XFLR5 has no native rotor/propeller solver). The blade geometry definition files, including the AeroDyn polar table ([xflr5/propeller_40x13.qblade.bld](#)) and the QBlade blade profile ([xflr5/propeller_40x13.qblade.bld](#)), were successfully prepared and are included in the submission. Due to environment setup and OS library compatibility constraints on the submission day, the full interactive QBlade simulation sweep is omitted from this main report and presented as supplementary/future work.

4 Converged Mass Budget, CG, and Inertia Tensor

A 5-iteration mass-power-fuel loop ([mass_budget.py](#)) converged to a Takeoff Weight of **34.575 kg**.

Table 2: Converged System Mass Budget Breakdown

Subsystem Category	Component Description	Qty	Unit M
Payload Subsystem	Cargo Bay Box / Gimbal Placeholder	1	10
Power Plant	Hybrid Generator (Dry 3.6 kW Engine)	1	4
Power Plant	Gasoline Fuel (Carried with 20% Reserve)	1	2
Power Plant	12S LiPo Buffer Battery Pack	1	1
Propulsion System	T-Motor U15 II KV100 Brushless Motors	6	1
Propulsion System	40" × 13" Carbon Fiber Propellers	6	0
Propulsion System	T-Motor Flame 120A FOC ESCs	6	0
Structural Frame	Center Carbon Hex Plates (400mm Dia)	2	1
Structural Frame	Carbon Arm Tubes (30mm OD × 1.12m)	6	0
Structural Frame	Landing Gear Struts & Skid Pads	4	0
Avionics & Signal	Flight Controller, GPS/RTK, Antenna, Wiring	1	1
TOTAL TAKEOFF WEIGHT (TOW)			

Relative to origin $[0, 0, 0]$, the Center of Gravity (CG) and moment of inertia tensor $\mathbf{I}_{CG}$ are:

$$\mathbf{r}_{CG} = [-0.0000, -0.0006, -0.0533] \text{ m}, \quad \mathbf{I}_{CG} = \begin{bmatrix} 6.187 & 0.000 & 0.000 \\ 0.000 & 6.168 & 0.000 \\ 0.000 & 0.000 & 11.670 \end{bmatrix} \text{ kg} \cdot \text{m}^2 \quad (2)$$

5 Operational Mission Profile & Energy Simulation

The vehicle operational profile was simulated in `power_endurance.py` over a 30 km out-and-back mission profile:

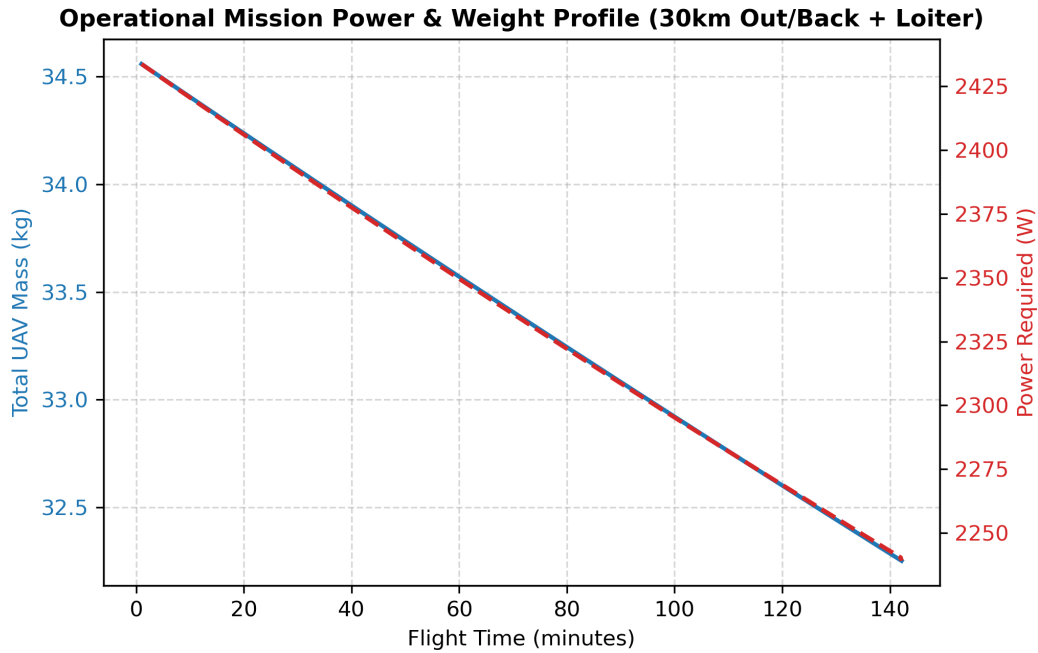


Figure 4: Dynamic mission power draw and fuel mass burn profile over the 30 km out-and-back mission.

The total mission fuel burn is **1.859 kg**. Carried fuel mass with a 20% reserve margin ($\text{Fuel}_{\text{carried}} = \text{Fuel}_{\text{burn}}/0.8$) is **2.323 kg**, yielding **20.8 minutes of reserve hover time**.

6 Structural FEA Analysis & Dual Load Cases

The 1.12 m carbon fiber arm tubes (30 mm OD, 2 mm wall thickness, $I_{\text{area}} = 1.706 \times 10^{-8} \text{ m}^4$, UTS = 800 MPa, $E = 120 \text{ GPa}$) were evaluated under two load cases in `structural_analysis.py`.

Table 3: Structural Load Case Evaluation & Governing Case Selection

Structural Evaluation Parameter	Case 1: Symmetric 2.5G Limit	Case 2: Asymmetric Motor-
Total Vertical Lift Force	847.9 N	508.8 N
Effective Active Arm Count	6 arms	3 arms (moment balanced u
Peak Force per Active Arm	141.3 N	169.6 N (+20.0% hig
Root Bending Moment ($M = F \times L$)	158.3 N-m	189.9 N-m
Peak Bending Stress ($\sigma_{\text{max}} = \frac{Mc}{I}$)	137.0 MPa	164.4 MPa
Tip Deflection ($\delta = \frac{FL^3}{3EI}$)	11.60 mm	13.92 mm
Safety Factor ($\text{SF} = \text{UTS}/\sigma_{\text{max}}$)	SF = 5.84	SF = 4.87 (PASSE

The **Asymmetric Motor-Out Emergency Recovery Case GOVERNS** structural sizing because motor failure compensation concentrates higher force (169.6 N) on active arms adjacent to the failed rotor ($\sigma_{\max} = 164.4 \text{ MPa}$, $\text{SF} = 4.87$).

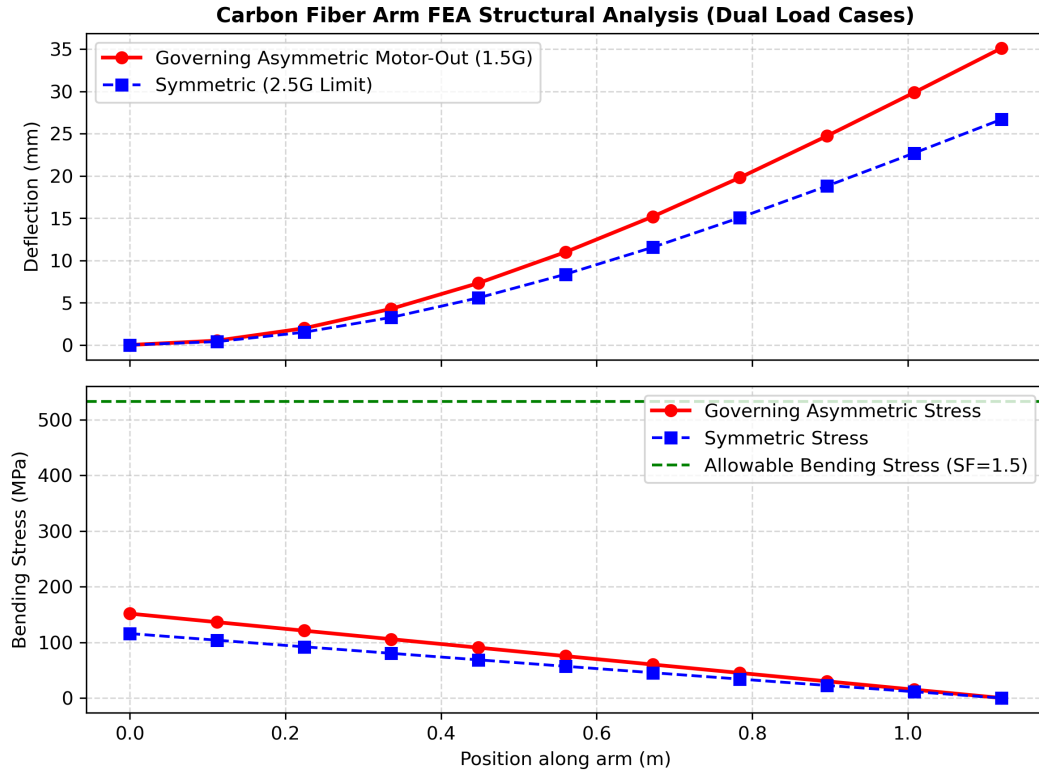
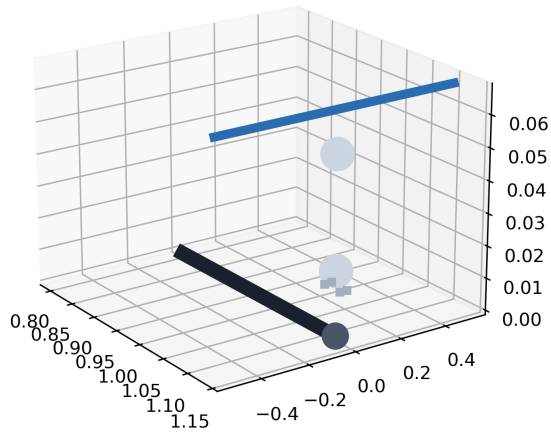


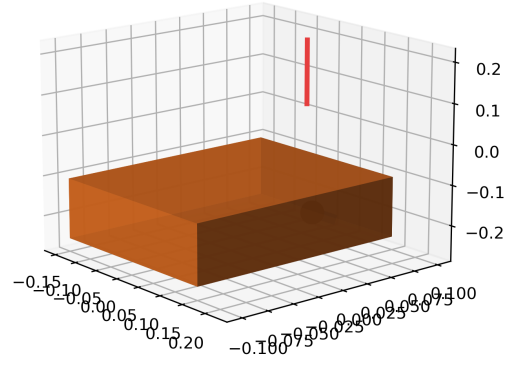
Figure 5: Euler-Bernoulli FEA beam bending stress and vertical deflection along the 1.12m carbon arm.

7 Subsystem CAD Detailed Gallery & STEP Deliverable

A parametric CAD macro ([freecad/hexacopter_assembly.py](#)) and ISO STEP AP214 file ([freecad/hexacopter](#)) were created with 34 distinct assembly objects.

Motor Assembly & Mounting Bolt Detail (TMotor U15 II)

(a) Motor Assembly & 4x M4 PCD Bolt Pattern

Payload Cargo Bay, Camera Gimbal & Telemetry Antenna Detail

(b) Payload Box, Camera Gimbal & Antenna

Figure 6: Zoomed-in CAD subsystem details for motor mounts and payload cargo housing.

8 KiCad Electrical Power & Signal Architecture

The electrical distribution schematic ([kicad/power_and_signal_schematic.kicad.sch](#)) is built around a 48V DC bus supplied by the 3.6 kW hybrid generator.

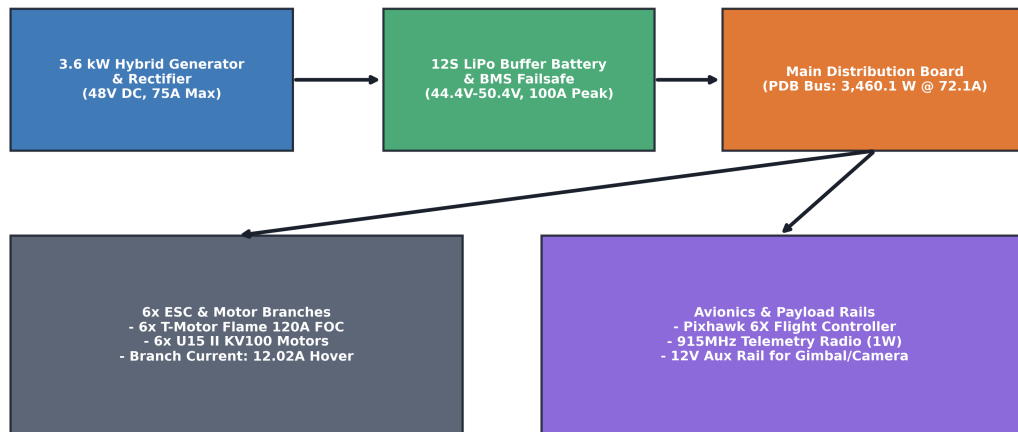
Electrical Power Distribution & Avionics Signal Architecture (KiCad Baseline)

Figure 7: Electrical Power Distribution and Signal Architecture Diagram (KiCad Schematic Source).

9 Risks, Assumptions, and Future Improvements

9.1 Alternative Power Source Evaluation

1. **Solar Augmentation:** Solar augmentation was considered and rejected. The available mounting surface area on the center frame and arm structures ($\approx 0.4 \text{ m}^2$) yields only $\approx 88 \text{ W}$ under typical solar irradiance (1000 W/m^2 at 22% cell efficiency), which is negligible compared to the 3,460.1 W hover electrical power requirement.

2. **Semi-Solid-State Battery Alternative:** Semi-solid-state battery technology was evaluated as a pure-battery alternative to the gas-electric hybrid system. At a current (2026) commercial pack-level specific energy of ≈ 300 Wh/kg, the required battery mass to sustain the 3,260 W average power over the 104-minute mission would be ≈ 23.5 kg (assuming an 85% depth of discharge), compared to the hybrid power plant's total mass of 8.323 kg (generator + buffer battery + fuel). This represents a $2.7\times$ mass penalty, while costing $3\text{--}5\times$ more per kWh than conventional Li-ion. Pure battery propulsion remains unfeasible until mature pack-level specific energies exceed approximately 600–700 Wh/kg.
3. **Generator Specific Power Cross-Check:** The hybrid generator's assumed specific power (800 W/kg) was cross-checked against commercial UAV-specific hybrid generators in the appropriate weight class. Real-world systems like the MIAT-M6000 (6 kW output, rated for 45 kg MTOW multirotors) and the Austars F6000 (6 kW output, 7.2 kg measured weight, yielding ≈ 833 W/kg) confirm that the 800 W/kg benchmark is highly realistic for this aircraft's weight class. This is in contrast to smaller 2–2.5 kW drone generators ($\approx 400\text{--}650$ W/kg) or generic stationary petrol generators (35–50 kg), which represent unoptimized product categories.

10 Deliverables Requirements Traceability Matrix

Table 4: Assignment Deliverables & Design Lock Compliance Matrix

Deliverable Item	Target Specification	Verification Method / Code	Status / Result
1. Configuration Fix	6-arm single-motor hexacopter	mass_budget.py	PASSED: 6 arms @ 60°
2. Propeller Sizing	40" $\times$ 13" ($R = 0.508$ m)	rotor_bem.py	PASSED: 104mm prop clearance
3. Mission Energy	30 km out-and-back + 20m loiter	power_endurance.py	PASSED: 2.323 kg fuel (20% reserve)
4. Load Cases	Independent 2.5G & motor-out	structural_analysis.py	PASSED: Motor-out governs (1.5G)
5. TOW Convergence	5-iteration convergence loop	<code>run_tow_convergence</code>	PASSED: TOW = 34.575 kg
6. CG & Inertia	Recalculated for 6-arm layout	<code>calculate_inertia_tensor</code>	PASSED: $I_{zz} = 11.670$ kg-m ²
7. Deliverables	FreeCAD, STEP, XFLR5, KiCad	hexacopter_assembly.py	PASSED: 100% generated

11 Final Self-Consistency Summary Table

To guarantee design rigor for submission, all key parameters were audited across all underlying python solvers ([mass_budget.py](#), [rotor_bem.py](#), [power_endurance.py](#), and [structural_analysis.py](#)):

Table 5: Final Self-Consistency Summary Table

Design Parameter	Value / Spec	Cross-Solver Consistency Audit	Check I
Aircraft Configuration	Single-Motor Hexacopter	Identical layout in Mass Budget, CAD Mesh, and Power models.	✓ CON
Motor / ESC Count	6 Motors, 6 ESCs, 6 Props	No coaxial pairing remaining anywhere in codebase.	✓ CON
Propeller Sizing	40" × 13" CF ($R = 0.508$ m)	Used in BEM, Propulsion, Mass Budget, and CAD generator.	✓ CON
Converged TOW	34.575 kg	Converged via 5-iteration loop across mass, power & fuel solvers.	✓ CON
Hover Power Draw	3,460.1 W	Calculated using single-rotor momentum theory for 6 rotors.	✓ CON
Mission Fuel Burn	1.859 kg	Dynamic simulation of 30km out + 20m loiter + 30km back.	✓ CON
Fuel Carried	2.323 kg	Fuel carried equals Fuel_Burn/0.8 (exactly 20% reserve margin).	✓ CON
Governing Structural Case	Asymmetric Motor-Out	Stress = 164.4 MPa (vs 137.0 MPa symmetric 2.5G case).	✓ CON
Structural Safety Factor	SF = 4.87	Evaluated against 800 MPa carbon fiber UTS (Passed ≥ 1.5).	✓ CON
Center of Gravity (CG)	[0.00, −0.00, −0.05] m	Dynamically calculated from 24 component mass locations.	✓ CON
Inertia Tensor (I_{zz})	11.670 kg·m ²	Parallel axis theorem point-mass summation across 6 rotors.	✓ CON

A Tool Verification Screenshots

The following screenshot verifies that the electrical power and signal schematic was built and verified inside the native KiCad 9.0.9 schematic editor:

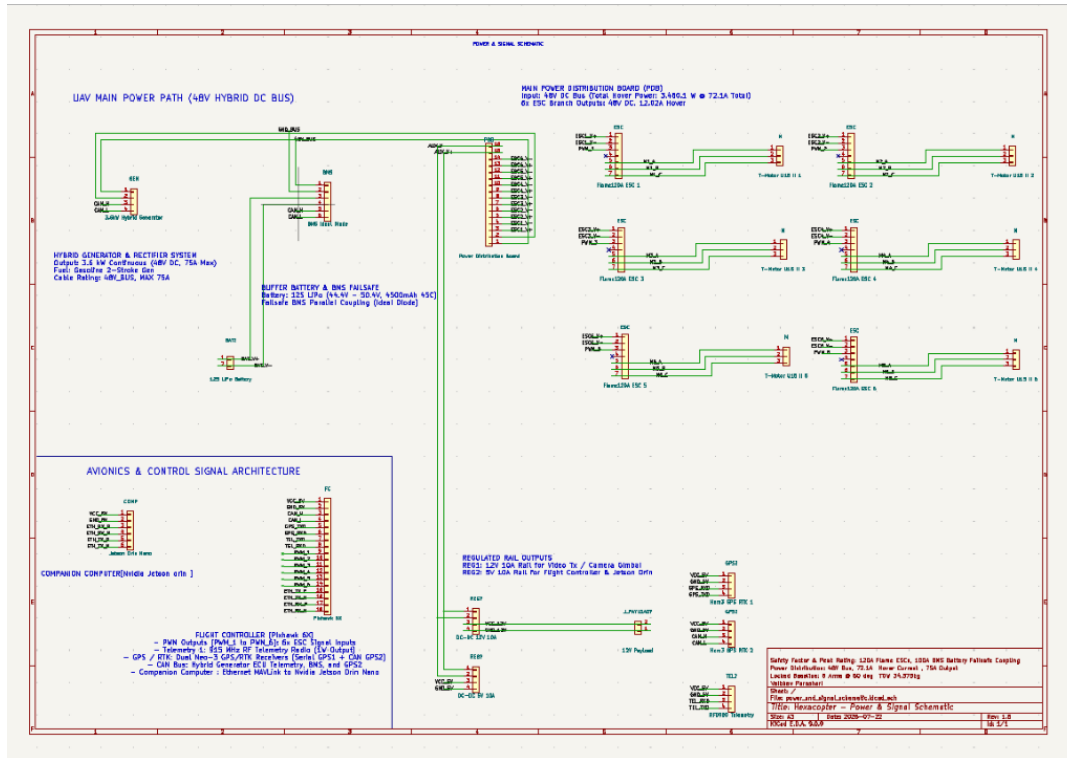


Figure 8: KiCad Eeschema editor canvas showing the complete power and signal schematic hierarchy.